

4.3L V6

Specification	4.3L TKS	4.3L MPI
Horsepower (performance obtained and corrected in accordance with SAE J1228)	190	220
Kilowatts (performance obtained and corrected in accordance with SAE J1228)	142	164
Displacement	4.3L (262 cid)	
Specified WOT RPM range (measured using an accurate service tachometer with the engine at normal operating temperatures)	4400–4800	4400–4800
Idle RPM in neutral (measured using an accurate service tachometer with the engine at normal operating temperatures)	600 (idle speed and timing on MPI models are not adjustable)	
Minimum oil pressure at idle (oil pressure must be checked with the engine at normal operating temperature)	55 kPa (4 psi)	
Minimum oil pressure at 2000 RPM (oil pressure must be checked with the engine at normal operating temperature)	124 kPa (18 psi)	
Thermostat	71° C (160° F)	
Thermostat (closed cooling)	76° C (170° F)	
Timing at idle (idle speed and timing on EFI models are not adjustable)	10° BTDC	Not adjustable
Firing order	1-6-5-4-3-2	
Electrical system	12 volt negative (–) ground	
Alternator rating (hot operating amps)	65 A	
Alternator rating (cold operating amps)	72 A	
Recommended battery rating for TKS models (minimum)	375 CCA, 475 MCA, or 90 Ah	
Recommended battery rating for MPI models (minimum)	750 CCA, 950 MCA, or 180 Ah	
Spark plug type	AC MR43LTS	NGK ITR4A15
Spark plug gap	1.14 mm (0.045 in.)	1.5 mm (0.060 in.)

Fluid Specifications

IMPORTANT: All capacities are approximate fluid measures.

Engine

IMPORTANT: It may be necessary to adjust oil levels depending on installation angle and cooling systems (heat exchanger and fluid lines).

Always use the dipstick to determine the exact quantity of oil or fluid required.

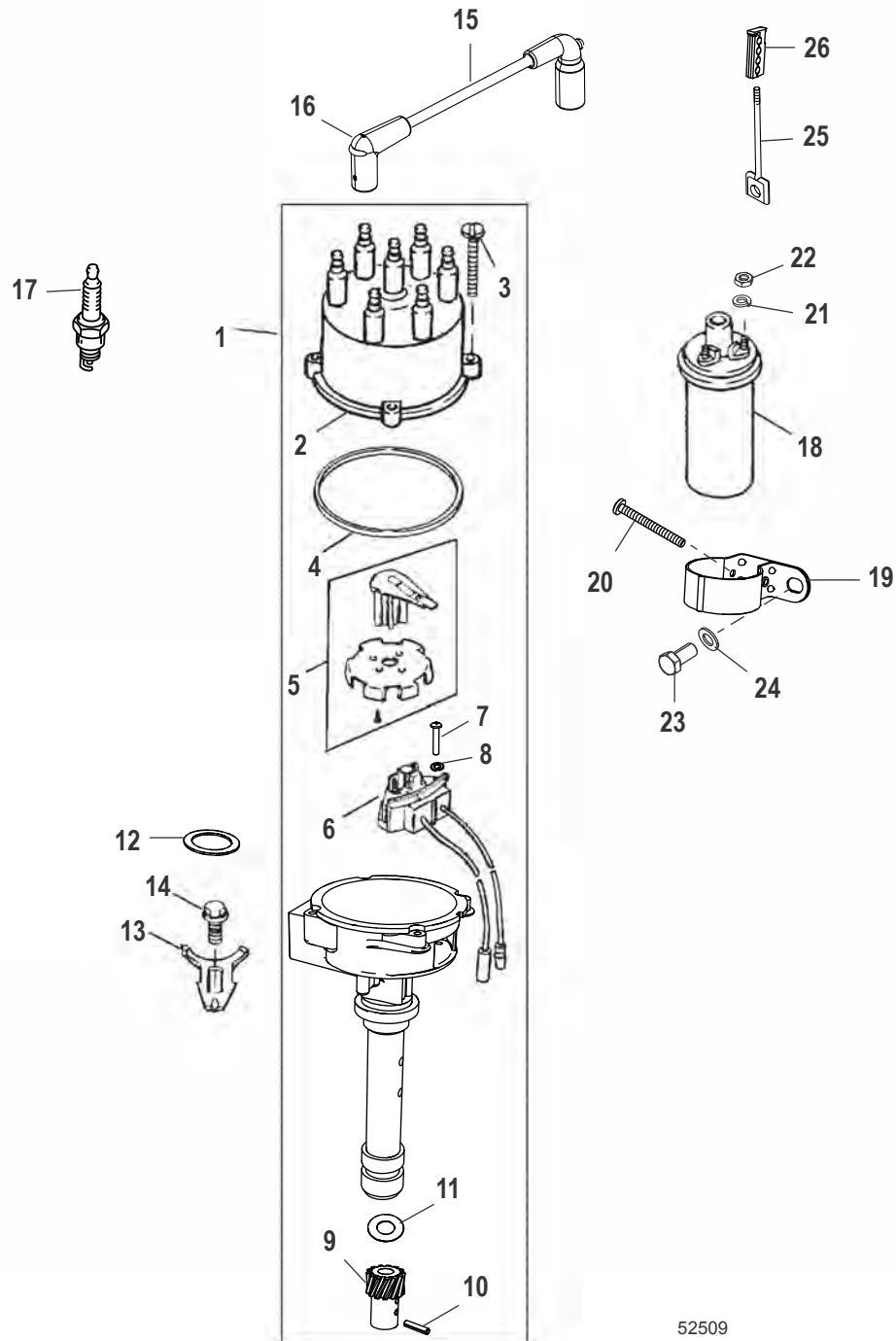
4.3L MPI and 4.3 TKS Models	Capacity	Fluid Type
Engine oil (with filter)	3.8 Liter (4 US qt)	Mercury/Quicksilver 25W-40 Synthetic Blend Engine Oil
Seawater cooling system (winterization use only)	20 Liter (21 US qt)	Propylene Glycol and Purified Water
Closed-cooling system	19 Liter (20 US qt)	Mercury Extended Life Coolant/Antifreeze or Extended Life Ethylene Glycol 5/100 Coolant/Antifreeze mixed 50/50 with Purified Water

Alpha Sterndrive

NOTE: Oil capacity includes gear lube monitor.

Model	Capacity	Fluid Type
Alpha One	1892 mL (64 oz)	High Performance Gear Lubricant

Distributor and Ignition Components (TKS)



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Distributor and Ignition Components (TKS)

Ref. No.	Qty.	Description		Torque		
				Nm	lb-in.	lb-ft
1	1	Distributor assembly				
2	1	Distributor cap				
3	4	Screw (#10-32 x 0.810)				
4	1	Gasket				
5	1	Rotor				
6	1	Sensor				
7	2	Screw (#10-32 x 1.25)				
8	2	Lockwasher				
9	1	Gear				
10	1	Pin				
11	1	Thrust washer				
12	1	Gasket				
13	1	Clamp				
14	1	Screw		34	–	25
15	1	Ignition wires				
16	1	Coil wire (NSS)				
17	6	Spark plug	New install	30	–	22
			Used install	15	133	–
18	1	Coil				
19	1	Bracket				
20	1	Screw				
21	2	Lockwasher (stainless steel)				
22	2	Nut (stainless steel)				
23	1	Screw (0.375-16 x 0.750)				
24	2	Washer (0.406 x 0.734 x 0.063)				
25	2	Stud				
26	2	Retainer				

Spark Plug Wire Removal

1. Disconnect the spark plug wires (high tension leads) from the spark plugs.

NOTE: Use care when removing the spark plug wires from the spark plugs. Twist the boot 1/2 turn before removing. Firmly grasp and pull on the boot, not the wire.

2. Remove the spark plugs.

NOTE: A thin-walled spark plug socket may be required.

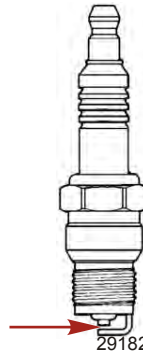
Spark Plug Wire Inspection

NOTE: Use care when removing the spark plug wires and the boots from the spark plugs. Twist the boot 1/2 turn before removing. Firmly grasp and pull on the boot to remove the wire.

1. Visually inspect the spark plug wires and the coil wire for damage.
2. Visually inspect the spark plug boots for damage.
3. Check the spark plug wires and the coil wire for continuity using a multimeter.
4. Replace any damaged wires.

Spark Plug Inspection

1. Inspect each spark plug for the manufacturer and spark plug number. All plugs must be from the same manufacturer and have the same spark plug number.



Spark plug gap

Engine	OEM Spark Plug	Mercury Part Number	Gap
Carbureted 4.3 TKS	AC 41-101	862029 AC MR43LTS	1.14 mm (0.045 in.)

2. Inspect each plug individually for worn electrodes and for glazed, broken, or blistered porcelain. Check the joint between the insulator and the shell for cracks. Replace as necessary.

Spark Plug Installation

1. Clean the cylinder head spark plug seat.
2. Adjust the spark plug gap with the appropriately sized round-tipped feeler gauge.

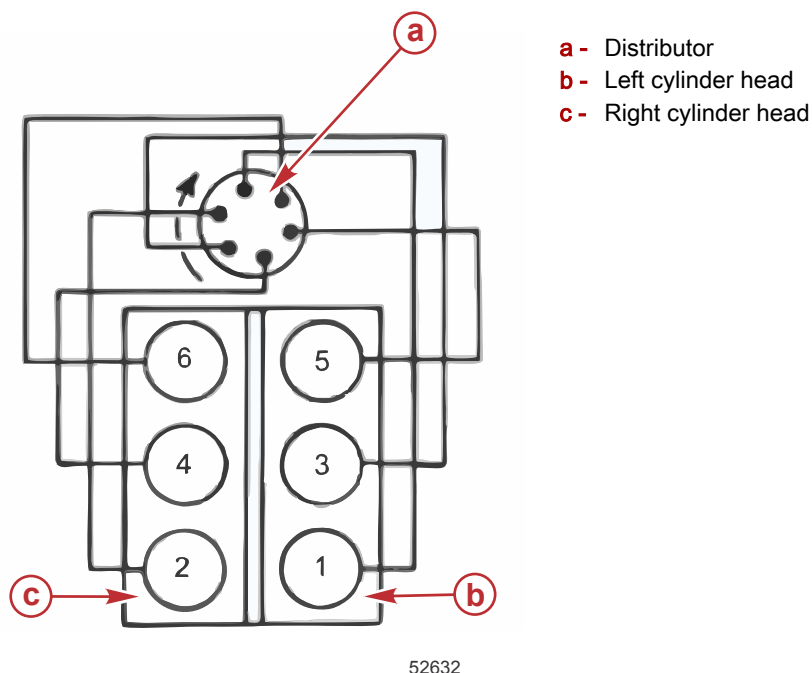
Engine	OEM Spark Plug	Mercury Part Number	Gap
Carbureted 4.3 TKS	AC 41-101	862029 AC MR43LTS	1.14 mm (0.045 in.)

IMPORTANT: In the absence of a torque wrench or if access to the plugs prevents torquing, hand-tighten the spark plugs until they seat in the cylinder head and then securely tightened with the appropriate ratchet and socket.

3. Install the spark plugs and tighten them to the specified torque.

Description	Nm	lb-in.	lb-ft
Spark plugs (new cylinder head)	30	–	22
Spark plugs (used cylinder head)	15	133	–

4. Install the spark plug wires in the proper order.



Description	Specification
Firing order	1-6-5-4-3-2

Thunderbolt V Models Timing

1. Connect the timing light to No. 1 spark plug wire.

Timing Light	91-99379
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2. Connect a shop tachometer to the engine.
IMPORTANT: Before starting the engine, connect a jumper wire from the ignition timing lead to a good ground. This has to be done before the ignition key is turned to the "ON" position to lock the ignition module into Base Timing Mode.
3. Before starting the engine, connect a jumper wire from the timing lead to a good ground.
NOTE: The purple/white timing lead is located towards the front of the engine near the fuel line or near the distributor, as equipped for your model.
4. Start the engine and operate it at 1300 RPM until it reaches normal operating temperature.
5. Disconnect the throttle cable from the carburetor.
6. With the engine at idle RPM, adjust the carburetor idle RPM screw to the specified engine idle RPM for your model.
7. With the engine still at idle RPM, check the ignition timing. If it is incorrect, rotate the distributor housing until the timing is correct.

Description	Nm	lb-in.	lb-ft
Distributor clamp bolt	24.5	–	18

IMPORTANT: The idle mixture screw is adjustable on models produced prior to January 1, 2003.

8. Check the idle RPM. If it is out of specification, adjust the idle mixture screw. Adjusting the idle mixture screw inward will lean the air-to-fuel ratio and adjusting it outward will enrich the air-to-fuel ratio.

Idle mixture adjustment tool	91-863375A1
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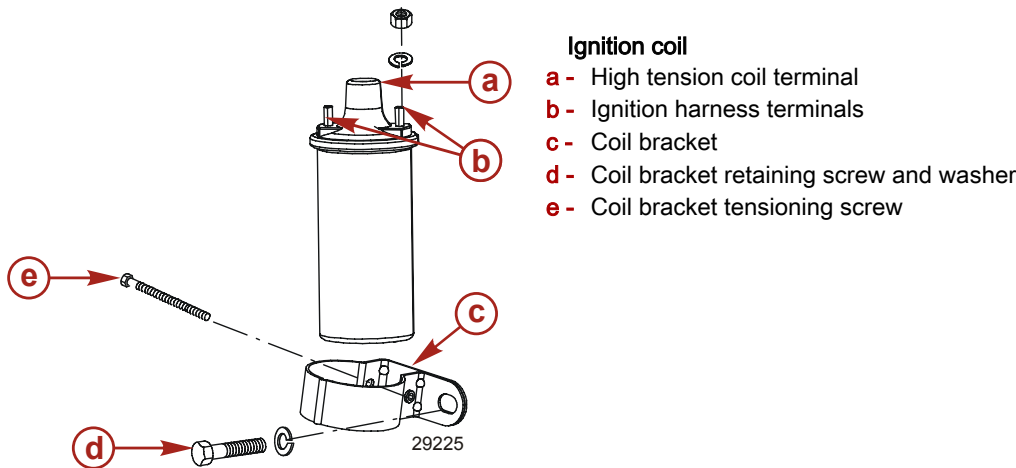
9. Recheck the ignition timing.
IMPORTANT: The timing jumper wire has to be removed or the ignition module will stay locked in the Base Timing Mode and it will not be able to advance the ignition timing correctly when the engine RPM is increased.
10. Stop the engine. Remove the timing light, jumper wire, and shop tachometer.
11. Reinstall and adjust the throttle cable. Open and close the remote control throttle lever to ensure that the carburetor's throttle lever returns against the RPM adjusting screw every time.

Ignition System

- 12. Restart the engine, increase the RPM to 1300, and return to idle position slowly. Ensure that the carburetor throttle lever returns against the idle RPM screw.
- 13. Shut the engine off.

Coil Removal

- 1. Disconnect the wire harness connectors at the ignition coil.

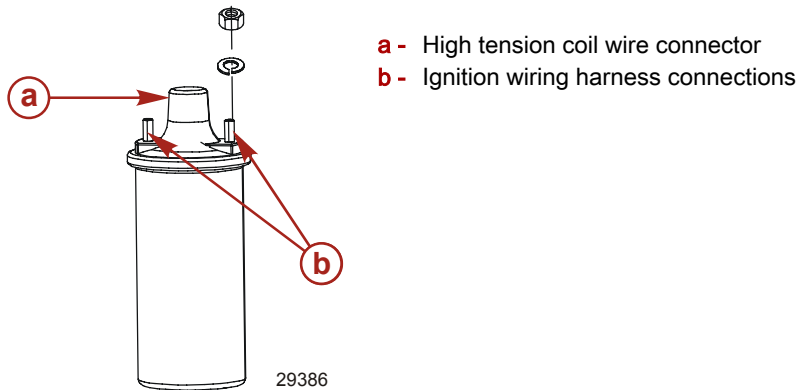


- 2. Remove the high tension coil lead.
- 3. Remove the coil bracket fasteners and bracket with the ignition coil.
- 4. Remove the ignition coil from the coil bracket.

Ignition Coil Testing

NOTE: Readings listed are taken at 20 °C (68 °F) with a digital ohmmeter. Due to the number of different manufactures of meters available and temperature variance, test results may vary.

- 1. Disconnect the ignition coil from the engine harness connector.
- 2. Remove the ignition coil high tension lead from the spark plug.



- 3. Check the resistance of the coil primary windings to engine ground. Replace the ignition coil if not within specification.

DMT 2004 Digital Multimeter		91-892647A01	
Meter Test Leads		Meter Scale	Reading (Ω)
Red	Black		
Coil primary lead	Coil mounting bolt	Auto range	0.17 - 0.23

- 4. Hold the high tension lead and rotate the spark plug cap counterclockwise. Remove the spark plug cap from the spark plug high tension wire.
- 5. Measure the resistance of the ignition coil secondary winding to engine ground. Replace the ignition coil if not within specification.

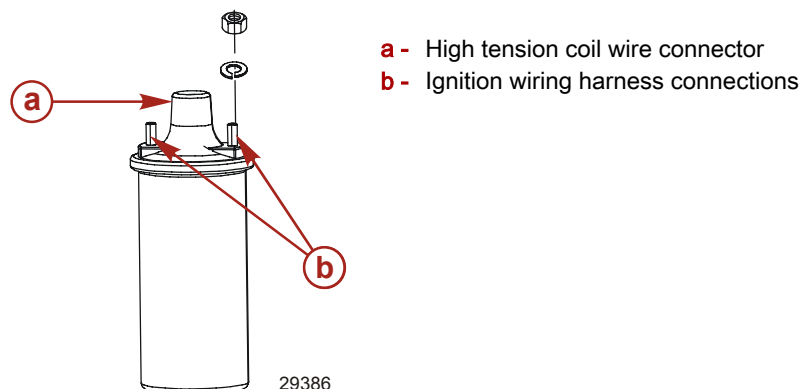
Meter Test Leads		Meter Scale	Reading (Ω)
Red	Black		
Coil high tension lead	Engine ground	Auto range	3.3 - 4.9k

Coil Installation

1. Install the ignition coil and bracket to the engine bracket.
IMPORTANT: Ensure that the sealant is not applied inside the coil high tension lead tower.
2. Apply a thin bead of sealant around the outside rim of the coil high tension lead tower. Ensure that the sealant is not applied inside the coil high tension lead tower.

Tube Ref No.	Description	Where Used	Part No.
 25 	Liquid Neoprene	Outside rim of the coil wire tower	92- 25711 3

3. Connect the high tension coil lead.
4. Connect the wire harness connectors.



Spark Control Features

Idle speed spark control—The ignition module controls ignition timing to maintain a calibrated idle speed by making small spark advance adjustments. This feature is only active between 400–700 RPM.

Acceleration spark advance—When accelerating, the ignition module may add more spark advance to the Base Spark Timing Curve. The amount of spark advance added depends on how fast RPM increases. This feature is active between 1200–4000 RPM. Within this range, the module can add approximately 10 degrees of spark advance to the base spark timing curve.

Mean-best-timing (MBT) spark advance—During light load cruising, the ignition module maintains optimal ignition timing by making small spark advance adjustments. At a given RPM, the module will add a small amount of advance and wait to see if there is an RPM change. If RPM increases, it will increase timing more. The module will continue to advance timing until it no longer gets an increase in RPM. Conversely, if it senses an RPM drop, it will start to retard some of the spark timing. Between 1200–4000 RPM the ignition module can add approximately 10–15 degrees of spark advance to the base spark timing curve.

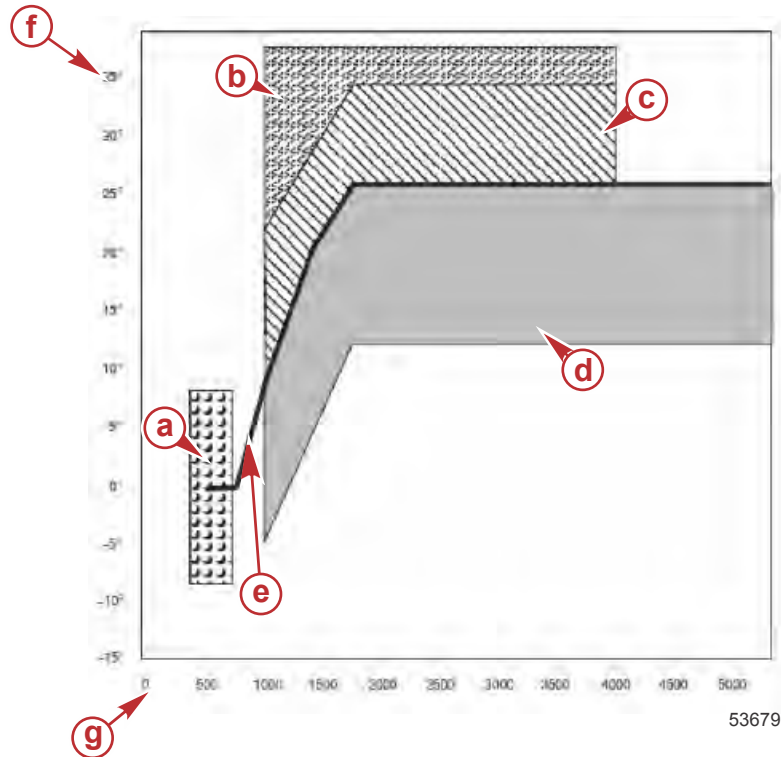
NOTE: The audio warning system is connected into the ignition module circuit. If the audio warning system becomes activated by the closing of one of the audio warning system switches, the MBT feature is deactivated.

Overspeed control—The ignition module will prevent the engine overspeed by adjusting the ignition spark signal. The overspeed limit is set slightly higher than the engine's recommended RPM range. Example: An operating RPM range of 4600–5000 RPM would have an overspeed limit of 5100 RPM and a reset RPM of 4750.

Knock retard spark control—The knock control module works in conjunction with the knock sensor and ignition module to retard ignition timing if knock is detected.

Thunderbolt V Spark Control Graph

IMPORTANT: The graph shows the typical advance ranges for a Thunderbolt V ignition control module. The numbers plotted on the graph are not representative of any particular model. It is only presented to provide an understanding of how the system functions.



Spark control graph

- a** - Idle speed advance range
- b** - MBT advance range
- c** - Acceleration advance range
- d** - Knock retard range
- e** - Base timing advance curve
- f** - Total spark advance minus initial timing
- g** - Engine RPM

Circuit Description

Refer to the **Ignition System Wiring Diagram** for reference to these circuit descriptions.

Ignition Control Module

- The ignition module receives power (+) through the purple wire.
- Ignition module ground (-) is accomplished through the black wire.
- There is also a case ground (-) wire that is connected to one of the ignition module mounting screws.
- The 12 volt signal from the ignition module to the distributor is carried through the white/red wire to the distributor sensor and back to the ignition module through the white/green wire.
- The tachometer signal is carried to the instrument panel through the gray wire.
- The purple/white wire carries the signal from the knock control module to the ignition control module.
- There are two black wires that have bullet connectors. This circuit is reserved for future options. On current models, the two black wires must be connected for the system to function properly.
- The tan/blue wire carries a signal from the audio warning circuit to the ignition module.

Knock Control Module

- The knock control module receives power (+) from the purple wire.
- Knock module ground (-) is accomplished through the black wire.
- The purple/white wire carries the signal from the knock control module to the ignition control module.
- The blue wire carries the signal from the knock sensor to the knock module.

Ignition Control System Timing Lead

The ignition control system has a lead with a bullet connector that is connected into the purple/white wire. When connected to an engine ground (-), this lead locks the ignition control module into the Base Timing Mode. This is used for performing the following tests and procedures:

- Setting the base ignition timing
- Setting the engine idle speed
- Setting the idle mixture
- Testing the knock control circuit

Ignition System Wiring Diagram

For 10-pin and 14-pin ignition system color wiring diagrams, refer to the **MerCruiser Color Diagram Binder**.

Ignition Module

1. Removal
 - a. Unplug the wiring harness connectors from the ignition module.
 - b. Remove the fasteners and hardware retaining the ignition module to the exhaust elbow.
 - c. Remove the module.
2. Ensure that the terminals of the wiring harness connector are clean and free of corrosion.
3. Installation
 - a. Install the ignition module using existing hardware. Tighten securely.
 - b. Plug the connectors into the ignition module.

Distributor Cap Removal

1. Loosen the distributor cap retaining screws.
2. Lift the cap away from the distributor.

Distributor Cap Cleaning and Inspection

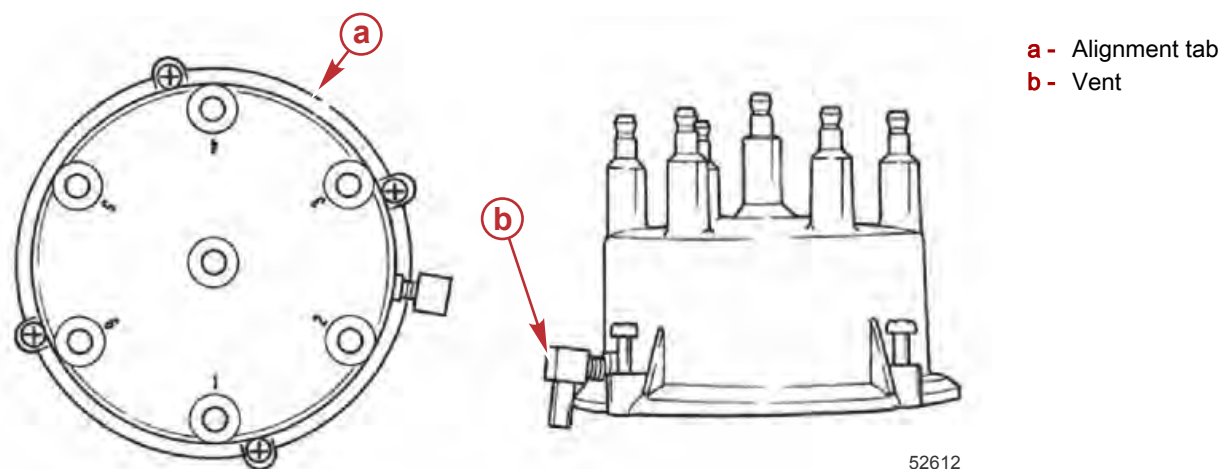
1. Check the cap contacts for excessive burning or corrosion.
2. Check the center contact for damage.
3. Check the cap for cracks or carbon tracks.
4. Clean the cap with soap and warm water. Dry it with compressed air.

Distributor Cap Installation

IMPORTANT: If the distributor cap is replaced, only use one recommended for marine applications.

NOTE: Use care when removing spark plug wires and boots from the distributor cap. Twist the plug wire boot 1/2 turn before removing it. Firmly grasp and pull on the boot to remove the wire.

1. Install the distributor cap by aligning the tab in distributor cap with the notch on the distributor body. Tighten the retaining screws.



2. For best results, individually transfer spark plug wires to the replacement cap in order of removal.

Rotor Assembly Removal

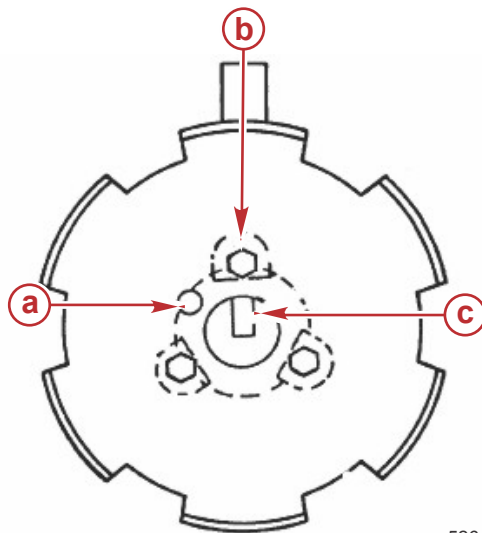
1. Position two flat blade screwdrivers opposite each other under the rotor with the blade tips contacting the distributor shaft.
2. Push down on both screwdriver handles at the same time to pry off the rotor assembly.

NOTE: The use of a torch lamp to heat the rotor assembly may aid in removal.

Torch Lamp	91-63209
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Rotor Assembly Cleaning and Inspection

1. Inspect the locating key inside the rotor for damage. The locating key should appear as a clean edged, 3 mm (1/8 in.) wide, sloped ramp at the bottom of the shaft hole.
2. Inspect the sensor alignment.
 - a. Set the sensor wheel on top of the following diagram with the sensor tangs facing up.



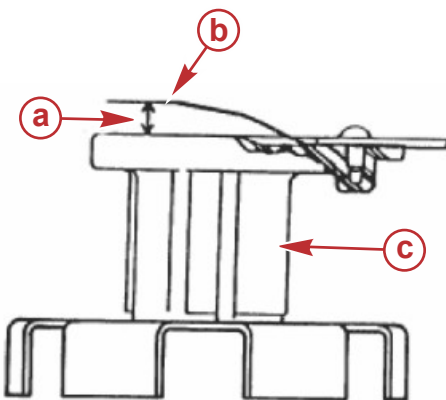
Rotor assembly sensor wheel diagram

- a - Locating pin
- b - Screws (3)
- c - Locating keyway

- b. Line up the three screw holes and locating pin hole on the sensor wheel with the diagram.
 - c. If the sensor wheel is indexed properly all the sensor fingers will line up with those in the diagram.
3. Check the rotor for a burned or corroded center contact or damaged key.
4. Check the rotor for cracks and carbon tracking.

Rotor Assembly Installation

1. Ensure that the carbon brush tang has a 6 mm (1/4 in.) gap between rotor and tang.



Rotor tang clearance

- a - Clearance measurement
- b - Carbon brush tang
- c - Rotor

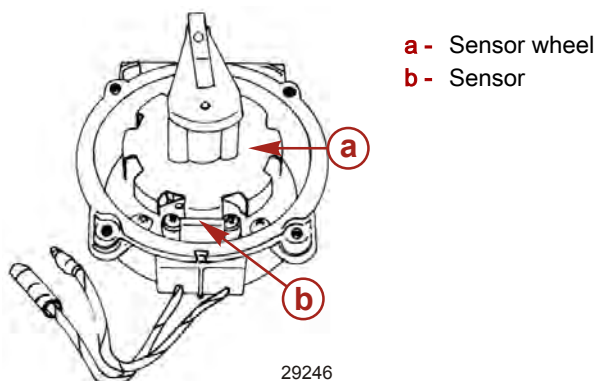
2. Put two drops of Loctite 271 Threadlocker into the rotor so it lands on the locating key.

Tube Ref No.	Description	Where Used	Part No.
 7	Loctite 271 Threadlocker	Key	92-809819

3. Put two drops of Loctite 271 Threadlocker in the keyway on the upper portion of the distributor shaft.
IMPORTANT: Do not let any Loctite run down the distributor shaft where it could contaminate the distributor housing bushing.

Tube Ref No.	Description	Where Used	Part No.
 7	Loctite 271 Threadlocker	Keyway	92-809819

4. Immediately install the rotor assembly on the distributor shaft. Ensure that the rotor locating key is aligned with the keyway in the distributor shaft. Press the rotor down on the shaft with your hand until it stops.
IMPORTANT: The rotor should fit tightly. It may be necessary to heat the rotor with a heat lamp to properly install it.
5. Let the Loctite 271 Threadlocker cure overnight with the distributor in an inverted position.



6. Install the distributor cap onto the distributor body.

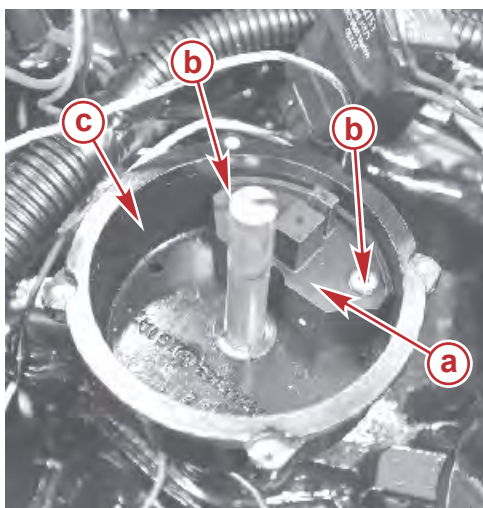
Sensor Testing

1. Disconnect the two sensor wires from the wiring harness.
2. Using an ohmmeter, check the resistance between the two sensor wires.
3. If the reading is less than 100 ohms, replace the sensor.

Description	Specification
Sensor resistance	>100 Ω

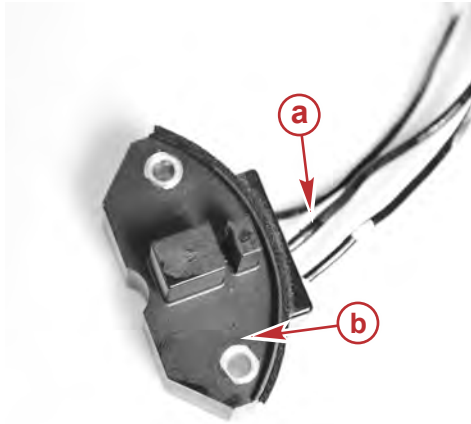
Sensor Removal and Inspection

1. Remove the rotor and sensor wheel.
2. Disconnect the sensor wires.
3. Remove the screws that hold the sensor in the distributor housing.
4. Remove the sensor from the housing.



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5. Use a magnifying glass and light to inspect the leads for cracks. If a crack is found in any lead, install a new sensor.

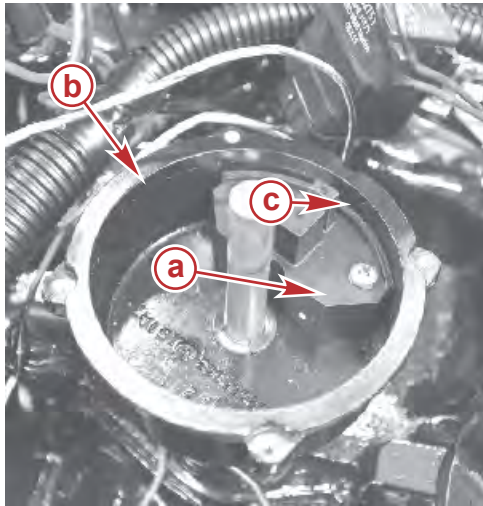


- a - Jumper leads
- b - Sensor assembly

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Sensor Installation

1. Install the sensor into the housing with the two retaining screws. Do not tighten them at this time.
2. Push the sensor outward until the rubber gasket is slightly compressed against the distributor housing. Then tighten the sensor mounting screws.



Distributor sensor adjustment

- a - Sensor
- b - Distributor housing
- c - Rubber gasket

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3. Install the sensor wheel, rotor, and distributor cap.
4. Connect the white/red and white/green sensor wires (ring terminals on engine harness wires, or bullet connectors from ignition module wires). Connect the black wire to engine ground. Tighten hex nuts securely (if equipped).

Distributor Removal

NOTE: One mounting screw protrudes through the distributor base and has a nut and washer. There is a ground lead attached to this screw.

1. Unplug the wiring harness from the ignition module. The ignition module will be mounted on the distributor housing or on the exhaust elbow, depending on the application.
2. Disconnect the sensor wires.
3. Remove the distributor cap. Do not remove the high tension leads unless necessary.
4. Crank the engine until the rotor is pointing toward the No. 1 cylinder on the distributor cap.
5. Make indexing reference marks on the distributor housing and engine block.
6. Remove the bolt and hold down clamp and remove the distributor.

IMPORTANT: Do not turn the engine while the distributor is removed.

Distributor Disassembly

1. Remove the rotor, sensor wheel, sensor, and ignition module (if it is mounted on the distributor).

2. Remove the roll pin, washer, and drive gear from the distributor shaft.
3. Check for side play between the shaft and distributor housing bushings.

Description	Specification
Distributor shaft to housing side play	0.05 mm (0.002 in.) maximum

4. Remove the shaft from the housing.
5. Check the distributor shaft runout with a dial indicator and V-blocks.

Description	Specification
Distributor shaft runout	0.05 mm (0.002 in.) maximum

Distributor Assembly

1. Lubricate the distributor shaft with SAE Engine Oil 30W.

Tube Ref No.	Description	Where Used	Part No.
 80	SAE Engine Oil 30W	Distributor shaft	Obtain Locally

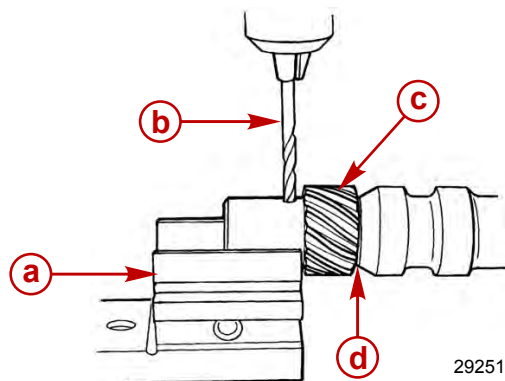
2. Install the E-clip onto the distributor shaft within the housing.
3. Install the washer onto the distributor shaft.
NOTE: The drive gear hole may be offset, allowing the drive gear to fit in only one direction.
4. Install the original drive gear. Slide it onto the distributor shaft and install the roll pin.

NOTE: A new drive gear will have a roll pin hole drilled on one side.

5. If installing a new drive gear:
 - a. Slide the drive gear onto the distributor shaft; align the hole in the gear with the hole in the distributor shaft.
 - b. Check the clearance between the drive gear and the distributor housing. The drive gear must turn freely.

Description	Specification
Distributor gear to distributor housing clearance	0.381 mm (0.015 in.)

- c. Using the hole in the drive gear and shaft as guides, drill through the other side of the drive gear with a 3/16 in. carbide tipped drill bit.



New distributor gear installation

- a - V-block
- b - 3/16 in. drill bit
- c - Drive gear
- d - Distributor gear to housing clearance

- d. Install the roll pin through the holes in the drive gear and distributor shaft.
- e. Recheck the clearance between the drive gear and the distributor housing. The drive gear must turn freely.

Description	Specification
Drive gear to distributor housing clearance	0.381 mm (0.015 in.)

- f. If the drive gear to distributor housing clearance does not meet specification or if the drive gear does not turn freely, rotate the drive gear 90° and redrill the roll pin holes.
6. Install the sensor, sensor wheel, rotor, and ignition module onto the distributor.

Distributor Installation

1. Position cylinder No. 1 at TDC (if not already done) by either of the following methods:
IMPORTANT: Do not insert anything into a spark plug hole while turning the engine over.

Ignition System

- a. Remove the No. 1 spark plug. Attach a suitable socket and drive to the crankshaft pulley nut. Place a finger over the plug hole and turn the engine over by hand until compression is felt in the No. 1 cylinder. Continue turning the engine over by hand until the block mounted timing pointer aligns with the timing mark on the crankshaft pulley.
 - b. Attach a suitable socket and drive to the crankshaft pulley nut. Remove the left (when observed from the rear of the engine) rocker cover and turn the engine over by hand until the No. 1 cylinder intake valve is fully closed. Continue turning the engine over by hand until the block mounted timing pointer aligns with the timing mark on the crankshaft pulley.
2. Position the distributor rotor so that it would fire cylinder No. 1. Use the distributor cap as a reference.
 3. Install a new gasket on the distributor housing.
IMPORTANT: Do not use the hold down clamp and bolt to force the distributor into position.
 4. Slide the distributor into the block until it stops on its own. Do not force the distributor.
IMPORTANT: Ensure that the oil pump shaft is properly engaged. If the engine has been turned over after disassembly it may be necessary to reposition the oil pump shaft.
 5. Raise the distributor up from its initial stopping point and rotate the rotor approximately 1/8 turn counterclockwise.
 6. Continue lowering the distributor into the block, gently working the distributor and rotor so that the oil pump shaft and camshaft gear are properly engaged. It may be necessary to rotate the rotor slightly until engagement is felt.
 7. When properly installed, the distributor housing is in full contact with the machined relief in the intake manifold.
 8. Connect all ignition control wiring.
 9. Install the distributor cap and gasket. Ensure that the rotor is still pointing at the No. 1 cylinder plug wire terminal.
 10. Coat any exposed wiring and terminals with sealant.

Tube Ref No.	Description	Where Used	Part No.
 25 	Liquid Neoprene	Electrical terminals	92- 25711 3